Lecture 4

Evaluation Scheme

(1) Quizzes: Approximately 2-3 times a month

(≈ 6 quizzes)

Mondays, 8:20 – 8:50 am.

#quizzes: q

Best (q-2) out of q quizzes.

40% of aggregate.

(2) Exams: Exam 1 – 10 Mar (Tue) } 6:30pm – 30%

Exam 2 – 30 Apr (Thu) } to 8:30pm – 30%

③ <u>Attendance:</u>

if # missed attendance calls $= m$

$$\text{Penalty} = \begin{cases} 0, & \text{if } m \leq 2 \\ 2(m-2), & \text{if } m > 2. \end{cases}$$

I. Periodic signals.

Defn: We say that a signal $x(t)$ is periodic if there exists a constant $T > 0$ such that,

$$x(t) = x(t+T).$$

Such a T is called <u>a period</u> of x.

See that: $x(t+2T) = x(\underline{t+T}+T)$

$$= x(t+T)$$

$$= x(t).$$

$\longrightarrow$ $2T$ is also a period.

$\Longrightarrow$ nT is a period of x for all $n \geq 1, 2, 3, \ldots$

Also: If $x(t+T) = x(t)$

Say $u = t+T$

$\Rightarrow x(u) = x(u-T)$

$\Rightarrow x(t) = x(t-T)$

$\Rightarrow$ we see that $x(t) = x(t-nT) \ \forall n \in \{1, 2, 3, \dots\}$.

$\therefore x(t) = x(t+nT) \ \forall n \in \mathbb{Z}$

Examples:

(1) Complex sinusoid.

Say $\omega > 0$. $\quad x(t) = e^{j\omega t} \quad \forall\, t.$

If $T > 0$ is a period: $\quad e^{j\omega t} = e^{j\omega(t+T)}$

$$\cancel{e^{j\omega t}} = \cancel{e^{j\omega t}} \cdot e^{j\omega T}$$

$$\Rightarrow e^{j\omega T} = 1$$

$$\Rightarrow \omega T = \text{an integer multiple of } 2\pi$$

$$\Rightarrow T = \frac{2\pi}{\omega} \times \text{positive integer.}$$

$\therefore$ smallest period is $T_0 = \dfrac{2\pi}{\omega}$.

(2) $x(t) = 1$ for all t.

Here, any $T > 0$ is a period of x.

This signal does not have a smallest possible period.

(because there is no smallest possible strictly positive real number).

<u>Defn</u>: If x has a smallest possible period (say T_0), then T_0 is called the "fundamental period" of x.

(3) $$x(t) = \begin{cases} 1, & \text{if } t \text{ is rational} \\ 0, & \text{otherwise} \end{cases}$$

Suppose T is rational number > 0.

$$\frac{p}{q} + T = \frac{p+qT}{q}$$

t	$t+T$
rational	rational
irrational	irrational

$\Longrightarrow x(t) = x(t+T)$.

Since there is no smallest positive rational, this signal does not have a fundamental period.

<u>Fact:</u> This particular signal x does not have a well-defined Riemann integration.

<u>Remark:</u> Unless otherwise mentioned we will assume that signals have a well-defined integration, and a well-defined fundamental period.

(4) Harmonically-Related family of complex sinusoids.
(Harmonics)

Suppose $\omega > 0$.

$$\phi_k(t) = e^{jk\omega t}, \quad k \in \mathbb{Z}.$$

is a complex sinusoid of frequency $\underline{k\omega}$ rad/s.
(recall that frequency can be -ve for complex sinusoids).

ϕ_k has fundamental period $\frac{2\pi}{|k|\omega}$ if $k \neq 0$

ϕ_0 : no fund. period, but any +ve number is a period.

Since $\frac{2\pi}{|k|\omega}$ is a period of ϕ_k $(k \neq 0)$

$\Rightarrow$ $\frac{2\pi}{|k|\cdot\omega} \times |k| = \frac{2\pi}{\omega}$ is also a period of ϕ_k.

$\Rightarrow$ all ϕ_k $(k \in \mathbb{Z})$ have a common period $= 2\pi/\omega$

Here ω is called the <u>fundamental freq.</u> <u>of the harmonics.</u>

II. Geometry of Periodic Signals.

(1) Energy of a Periodic signal.

Suppose x is T-periodic (T is a period of x).

$$E = \int_0^T |x(t)|^2 \, dt$$

(We fix T to be any valid period; most often, T is chosen as the fundamental period)

(2) Power of a T-periodic signal

$$P = \frac{E}{T}$$

③ <u>Innex Product</u>

If T is a period of both x and y.

$$(x, y) = \int_0^T x(t)\, y^*(t)\, dt$$

$(\langle x, y \rangle)$

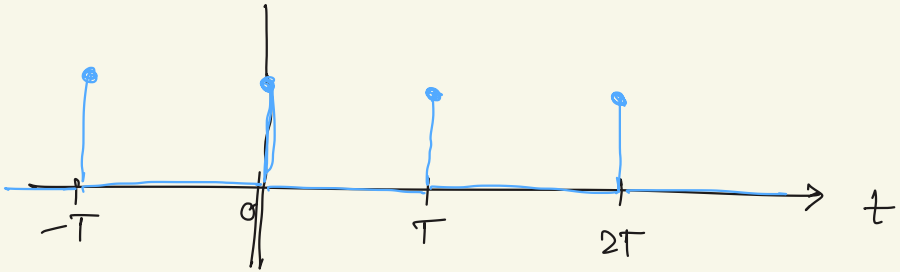
(i) $$(x, x) = \int_0^T x(t) \cdot x^*(t)\, dt = E$$

<u>Notation:</u> The norm of x is defined as:

$$\| x \| = \sqrt{(x, x)} \geq 0 \qquad (\text{that is } \| x \| = \sqrt{E}).$$

This (norm) is a measure of how far x is from the all-zero signal

$\|x\|$ (or $\|x\|_2$) is the 2-norm or Euclidean norm of x.



$$x(t) = \begin{cases} 1 & \text{if } t = kT, \ k \in \mathbb{Z} \\ 0 & \text{otherwise} \end{cases}$$

For this x, we have $\|x\| = 0$

$\therefore \ \|x\| = 0 \not\Rightarrow x(t) = 0 \ \forall t.$

Properties of inner product: (T is fixed)

1. $(x, x) \geq 0$

   ($x = 0 \Rightarrow (x, x) = 0$. converse not true).

2. $(x, y) = (y, x)^*$

3. $(\alpha x, y) = \int_0^T \alpha x(t) \cdot y^*(t)\, dt = \alpha (x, y)$

   $(x, \alpha y) = \int_0^T x(t) \cdot [\alpha y(t)]^*\, dt = \alpha^* (x, y)$.

4. $(x_1 + x_2, y) = (x_1, y) + (x_2, y)$

   $(x, y_1 + y_2) = (x, y_1) + (x, y_2)$.